PharmaLeaf · B.Pharm Semester II · PCI 2026 (NEP 2020)

BP201T Applied Biostatistics and Data Analytics for Pharmaceutical Sciences

PharmaLeaf notes for B.Pharm Semester II (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order, with tables, diagrams, hand-worked examples and Python programs. Every program in these notes was run, and the output shown is the real output.

Syllabus at a glance

BP201T is a 2-credit core theory course: 2 lectures per week, 30 hours in total, 50 marks (20 sessional + 30 end-semester).

Unit	Title	Topics	Hours
I	Descriptive statistics	Types and sources of data; mean, median, mode; range, variance, SD; skewness; descriptive statistics in Python	6
II	Probability and statistical distributions in healthcare	Probability laws; conditional probability; Bayes' theorem; random variables; normal, binomial and Poisson distributions; plotting in Python	6
III	Sampling and statistical inference	Population and sample; sampling techniques; sampling error and bias; Central Limit Theorem; confidence intervals; hypothesis testing; Type I and II errors; p-value	6
IV	Basics of correlation and regression	Pearson correlation; positive and negative correlation; scatter plots; simple linear regression; odds ratio	6
V	Statistical analysis using Python: case-based learning	SciPy, Statsmodels, Scikit-learn on pharmaceutical data; reading output and p-values; writing a statistical report	6

How to use these notes

- Each topic gives the definition first (for 2-mark answers), then formulas, a worked example done by hand and a table or diagram (for 5- and 10-mark answers).
- Every program can be typed and run in Jupyter Notebook, Google Colab, PyCharm or VS Code. The box labelled Output shows exactly what the program prints.

- The four practice files (`tablet_weights.csv`, `bp_trial.csv`, `dose_response.csv`, `calibration.csv`) are given in full in the Appendix at the end. Keep them in a folder named `data` next to your notebook.

Unit I: Descriptive Statistics (6 hours)

1.1 Statistics and biostatistics

Statistics: the science of collecting, organising, summarising, analysing and interpreting numerical data to draw conclusions.

Biostatistics: the application of statistics to biology, medicine and pharmacy, for example in clinical trials, quality control and epidemiology.

Two branches of statistics

Branch	What it does	Pharmacy example
Descriptive statistics	Summarises the data in hand using tables, graphs and numbers (mean, SD)	Average weight and SD of 20 tablets from a batch
Inferential statistics	Draws conclusions about a whole population from a sample, with a measure of uncertainty	Deciding from 200 patients whether a new drug lowers BP in all hypertensive patients

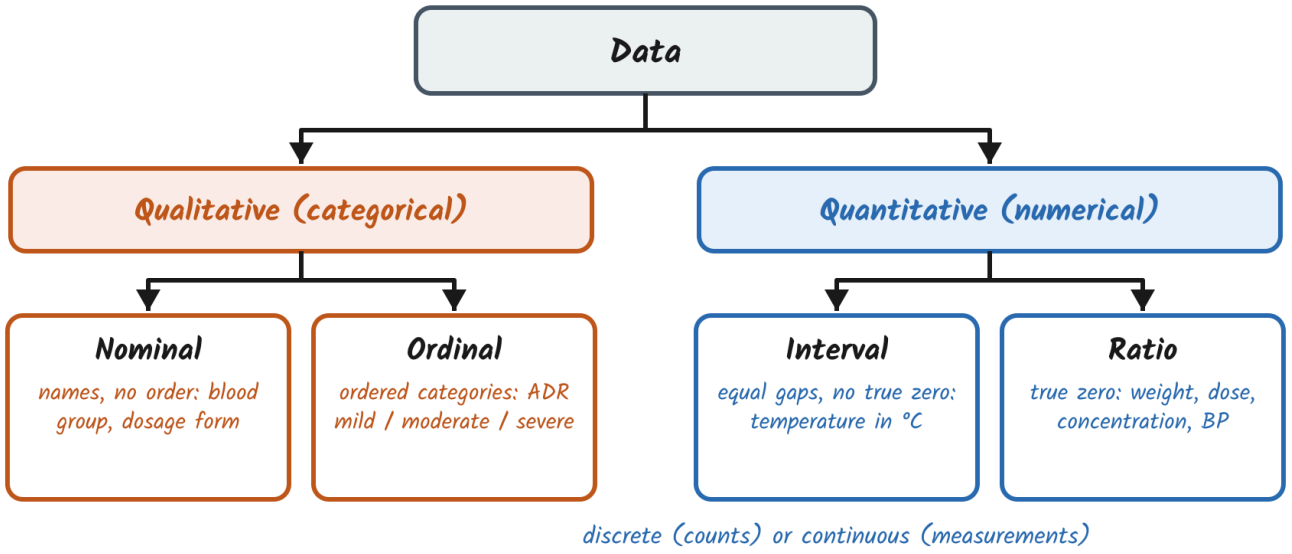
Uses of biostatistics in pharmacy

1. Designing clinical trials and calculating sample size.
2. Comparing a new drug with a placebo or a standard drug.
3. Quality control: weight variation, content uniformity, process control charts.
4. Pharmacokinetics: calculating AUC, half-life and bioequivalence.
5. Pharmacovigilance: detecting signals of adverse drug reactions (ADRs).
6. Stability studies: predicting shelf life.
7. Epidemiology: finding risk factors of disease (odds ratio, relative risk).

1.2 Types of data in pharmaceutical sciences

Data: a set of recorded observations or measurements. A characteristic that changes from one subject to another (age, weight, blood group) is called a **variable**.

Types of Data (NOIR)



Remember: N-O-I-R: each level adds one property
Order (ordinal) -> equal gaps (interval) -> true zero (ratio). Data type decides the average and the test.

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Type (scale)	Meaning	Can you order it?	Equal intervals?	True zero?	Pharmacy examples
Nominal	Names or categories with no order	No	No	No	Gender, blood group, dosage form (tablet, syrup), route of administration
Ordinal	Categories with a natural order, but gaps are not equal	Yes	No	No	ADR severity (mild, moderate, severe), pain score, disease stage I-IV
Interval	Numbers with equal gaps, but no true zero; ratios are meaningless	Yes	Yes	No	Temperature in °C, calendar year
Ratio	Numbers with equal gaps and a true zero; ratios are meaningful	Yes	Yes	Yes	Tablet weight, drug concentration, dose, age, BP, heart rate

Qualitative vs quantitative data

- **Qualitative (categorical):** describes a quality; nominal and ordinal data.
- **Quantitative (numerical):** measured or counted numbers; interval and ratio data.